

The General Structure of Trilinear Equations

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Abstract

We investigate trilinear structures as a natural extension of the Hirota bilinear formalism in integrable systems. While bilinear equations are associated with Grassmannian geometry and Plücker relations, trilinear equations suggest a higher algebraic structure involving three-slot couplings of tau functions.

Focusing on the stationary axisymmetric Einstein equations, we show that when the Ernst potential is written in a tau-ratio form, the nonlinear equation decomposes into a cubic sector containing all second-derivative terms and a quartic gradient envelope. The cubic sector is identified with a YTSF-type trilinear kernel.

We formulate a general trilinear kernel criterion and apply it to the Tomimatsu–Sato solutions. In particular, we demonstrate that the $\delta = 3$ solution possesses the same trilinear kernel structure as the $\delta = 2$ case, with a universal normalization up to a constant factor.

These results suggest that the trilinear kernel represents a universal structure governing the highest-derivative sector of the Ernst system, providing a new perspective on integrability beyond the bilinear hierarchy.

1 Introduction

The Hirota bilinear formalism plays a central role in the theory of integrable systems [1, 2]. It provides a powerful method for rewriting nonlinear partial differential equations in a bilinear form, and is deeply connected with the Grassmannian structure of Sato theory [3, 4].

It is therefore natural to ask whether there exists a higher multilinear structure beyond the bilinear level, and if so, whether such a structure possesses a comparable universality.

In this paper we investigate trilinear structures as a candidate for such an extension. In particular, we focus on nonlinear systems that admit a tau-function representation and examine the structure of their highest-derivative terms.

A key observation arises in the stationary axisymmetric Einstein equations. When the Ernst potential [5] is written in a tau-ratio form, the nonlinear equation naturally decomposes into a cubic part and a quartic gradient envelope. Remarkably, all second-derivative terms are contained in the cubic sector, suggesting the presence of a trilinear kernel governing the dynamics [6, 7].

Motivated by this observation, we formulate a general framework for identifying trilinear integrable kernels. We introduce a Z_3 -symmetric trilinear Hirota operator, define a kernel criterion, and apply it to the Tomimatsu–Sato solutions [8]. In particular, we demonstrate that the $\delta = 3$ case exhibits the same trilinear kernel structure as the $\delta = 2$ solution.

We further discuss the possible universality of this structure within the Tomimatsu–Sato hierarchy and its physical interpretation in terms of second-order nonlinear dynamics.

The remainder of this paper is organized as follows. After introducing the general structure of trilinear equations in Sec. 2, we establish the trilinear kernel criterion in Sec. 3. The $\delta = 3$ Tomimatsu–Sato solution is then examined in Sec. 4, followed by a discussion of the universality of the trilinear kernel within the TS hierarchy in Sec. 5. Finally, Sec. 6 provides a physical

interpretation of the transition from bilinear to trilinear dynamics. Technical details of the coefficient analysis and its computational verification are presented in Appendices A and B.

2 The General Structure of Trilinear Equations

The purpose of this section is to formulate the basic viewpoint underlying trilinear equations. In the conventional theory of soliton equations, Hirota bilinear equations occupy a distinguished position. They are not merely a convenient rewriting of nonlinear partial differential equations, but reflect the Pluecker relations of the Grassmannian and hence possess a universal geometrical origin. It is therefore natural to ask whether trilinear equations, such as those appearing in the Yu–Toda–Sasa–Fukuyama (YTSE) system [6, 7], also admit a comparable universal structure.

Let

$$\omega = \exp\left(\frac{2\pi i}{3}\right), \quad 1 + \omega + \omega^2 = 0. \quad (1)$$

For three functions f, g, h , we define the elementary trilinear Hirota operator by

$$T_x(f, g, h) = (\partial_{x_1} + \omega\partial_{x_2} + \omega^2\partial_{x_3}) f(x_1)g(x_2)h(x_3) \Big|_{x_1=x_2=x_3=x}. \quad (2)$$

This operator is characterized by a Z_3 -symmetric phase structure. In particular,

$$T_x(f, f, f) = 0, \quad (3)$$

which is the trilinear analogue of the cancellation property familiar in Hirota’s bilinear formalism.

A general trilinear equation may then be regarded as a nonlinear equation whose essential differential part is organized by such three-slot operators. In contrast to bilinear equations, which involve two copies of a tau function, trilinear equations involve three tau slots and therefore describe a genuinely three-body algebraic coupling among tau functions:

$$(f, g, h) \longmapsto T(f, g, h). \quad (4)$$

The word “trilinear” should not be interpreted as referring to three spatial dimensions. Rather, it refers to the algebraic multiplicity of the tau functions entering the nonlinear structure.

In the simplest two-component setting one considers a pair of tau functions

$$\tau_0, \quad \tau_1, \quad (5)$$

and constructs the basic YTSE-type kernel [7]

$$\begin{aligned} \mathcal{Y}(\tau_0, \tau_1) = & \partial_x [\tau_1 T_x(\tau_0, \tau_0, \tau_1) - \tau_0 T_x(\tau_1, \tau_1, \tau_0)] \\ & + \partial_y [\tau_1 T_y(\tau_0, \tau_0, \tau_1) - \tau_0 T_y(\tau_1, \tau_1, \tau_0)]. \end{aligned} \quad (6)$$

This expression is cubic in the tau functions and their derivatives. It contains terms of the schematic form

$$\tau_0^2 \partial^2 \tau_1 - \tau_1^2 \partial^2 \tau_0 + \text{lower derivative terms}. \quad (7)$$

Thus the highest-derivative sector is governed by a trilinear tau-function structure.

This observation suggests the following working definition. A nonlinear equation is said to possess a trilinear integrable kernel if, after an appropriate tau-function representation, its highest-derivative sector can be written in terms of a Z_3 -symmetric trilinear Hirota operator, while the remaining terms contain only lower derivatives. Symbolically,

$$\mathcal{N}(\tau) = \mathcal{N}_{\text{tri}}(\tau) + \mathcal{N}_{\text{lower}}(\tau), \quad (8)$$

where

$$\mathcal{N}_{\text{tri}}(\tau) \sim \mathcal{Y}(\tau_0, \tau_1), \quad (9)$$

and $\mathcal{N}_{\text{lower}}$ contains no terms of the highest differential order.

From this point of view, trilinear equations form a natural extension of the bilinear Hirota formalism. The bilinear case is associated with pairwise relations among tau functions, determinant identities, and Pluecker relations. The trilinear case may instead be viewed as encoding a higher algebraic interaction among tau functions. Whether this structure is governed by a higher analogue of the Pluecker relations, by a flag-type geometry, or by another universal algebraic principle remains an open problem.

The stationary axisymmetric Einstein equations provide a particularly interesting realization of this idea. If the Ernst potential is written as a tau ratio [5],

$$\mathcal{E} = \frac{\tau_1}{\tau_0}, \quad (10)$$

then the numerator of the Ernst equation naturally separates into a cubic part and a quartic gradient envelope,

$$\mathcal{N} = \mathcal{N}_{\text{cubic}} + \mathcal{N}_{\text{quartic}}. \quad (11)$$

The crucial point is that all second-derivative terms belong to $\mathcal{N}_{\text{cubic}}$, while $\mathcal{N}_{\text{quartic}}$ contains only first-derivative products. Therefore the dynamical kernel of the equation is cubic, not quartic. In the Tomimatsu–Sato $\delta = 2$ spacetime [8] this cubic sector is precisely of YTSF trilinear type [7].

This suggests a general physical interpretation. Bilinear equations describe integrable dynamics organized by pairwise tau-function relations. Trilinear equations describe integrable dynamics in which the fundamental nonlinear kernel is a three-slot tau-function coupling. In this sense the YTSF structure may represent a genuine next layer of integrability beyond the conventional bilinear hierarchy.

Several questions then become natural:

1. What is the universal geometrical origin of trilinear Hirota equations?
2. Is there a trilinear analogue of the Pluecker relation?
3. Can one construct a hierarchy of trilinear equations analogous to the KP or Toda hierarchy [9–11] ?
4. Do quartic or higher multilinear equations possess independent integrable meaning, or are they reducible to bilinear and trilinear kernels plus lower-order envelopes?
5. In gravitational systems, is the trilinear kernel a special property of the Tomimatsu–Sato family, or a more general feature of stationary axisymmetric vacuum Einstein equations?

The central proposal is therefore that trilinear equations should be studied not as exceptional algebraic curiosities, but as candidates for a second universal integrable structure, standing next to the bilinear Grassmannian structure of Sato theory.

3 Trilinear Kernel Test in the Ernst System

In this section we formulate a general criterion to test whether a given stationary axisymmetric solution of the vacuum Einstein equations admits a trilinear integrable kernel in the sense introduced in Sec. 2.

We start from the Ernst equation

$$\mathcal{E} \Delta \mathcal{E} - (\nabla \mathcal{E})^2 = 0, \quad (12)$$

and assume that the Ernst potential can be written in a tau-ratio form

$$\mathcal{E} = \frac{\tau_1}{\tau_0}, \quad (13)$$

with sufficiently smooth functions τ_0, τ_1 .

Substituting this form into the Ernst equation and clearing denominators, we obtain

$$\mathcal{E} \Delta \mathcal{E} - (\nabla \mathcal{E})^2 = \frac{\mathcal{N}(\tau_0, \tau_1)}{\tau_0^4}, \quad (14)$$

where $\mathcal{N}$ is a polynomial in τ_0, τ_1 and their derivatives. As discussed in Sec. 2, this numerator admits a natural decomposition

$$\mathcal{N} = \mathcal{N}_{\text{cubic}} + \mathcal{N}_{\text{quartic}}, \quad (15)$$

with

$$\mathcal{N}_{\text{cubic}} = \tau_1 \tau_0^2 \Delta \tau_1 - \tau_1^2 \tau_0 \Delta \tau_0, \quad (16)$$

$$\mathcal{N}_{\text{quartic}} = -\tau_0^2 |\nabla \tau_1|^2 + \tau_1^2 |\nabla \tau_0|^2. \quad (17)$$

The crucial point is that all second-derivative terms are contained in $\mathcal{N}_{\text{cubic}}$, while $\mathcal{N}_{\text{quartic}}$ contains only first derivatives.

We now introduce the YTSF-type trilinear kernel $\mathcal{Y}(\tau_0, \tau_1)$ defined in Sec. 2. To test whether the given system possesses a trilinear integrable kernel, we consider the difference

$$\mathcal{Q}(\tau_0, \tau_1) = \mathcal{N}_{\text{cubic}} - \kappa \mathcal{Y}(\tau_0, \tau_1), \quad (18)$$

with a constant κ to be determined.

We then adopt the following criterion:

Trilinear kernel criterion. A stationary axisymmetric solution is said to possess a trilinear integrable kernel if there exists a constant κ such that

$$\mathcal{Q}(\tau_0, \tau_1) \quad (19)$$

contains no second derivatives of τ_0, τ_1 .

Equivalently, the projection of $\mathcal{Q}$ onto the space spanned by $\partial^2 \tau_0$ and $\partial^2 \tau_1$ vanishes identically. In that case the entire second-derivative structure of the Ernst equation is governed by the trilinear kernel $\mathcal{Y}$.

This formulation is independent of the explicit form of the solution and provides a universal test applicable to a wide class of stationary axisymmetric spacetimes.

4 The Tomimatsu–Sato $\delta = 3$ Case

We now apply the trilinear kernel criterion to the Tomimatsu–Sato spacetime with multipole parameter $\delta = 3$. In the correspondence with the TS solutions, the multipole parameter δ is identified with the size of the determinant [9],

$$N = \delta. \quad (20)$$

Therefore, the $\delta = 3$ TS solution is described by 3×3 determinant tau functions. Thus, the Ernst potential is written in tau-ratio form as

$$\mathcal{E}_{(3)}(\xi, \eta) = \frac{\tau_1^{(3)}(\xi, \eta)}{\tau_0^{(3)}(\xi, \eta)}. \quad (21)$$

Here $\tau_0^{(3)}$ and $\tau_1^{(3)}$ are the rank-three Toda-molecule tau functions. In determinant form they may be represented as

$$\tau_n^{(3)} = \det (m_{i+j+n})_{i,j=0,1,2}, \quad n = 0, 1, \quad (22)$$

where the moments $m_k(\xi, \eta)$ are those associated with the $\delta = 3$ TS solution. Thus explicitly

$$\tau_0^{(3)} = \begin{vmatrix} m_0 & m_1 & m_2 \\ m_1 & m_2 & m_3 \\ m_2 & m_3 & m_4 \end{vmatrix}, \quad \tau_1^{(3)} = \begin{vmatrix} m_1 & m_2 & m_3 \\ m_2 & m_3 & m_4 \\ m_3 & m_4 & m_5 \end{vmatrix}. \quad (23)$$

Substituting

$$\tau_0 = \tau_0^{(3)}, \quad \tau_1 = \tau_1^{(3)}$$

into the cubic Ernst numerator, we define

$$\mathcal{N}_{\text{cubic}}^{(3)} = \tau_1^{(3)} \left(\tau_0^{(3)} \right)^2 \Delta \tau_1^{(3)} - \left(\tau_1^{(3)} \right)^2 \tau_0^{(3)} \Delta \tau_0^{(3)}. \quad (24)$$

We then compare this expression with the YTSF-type trilinear kernel $\mathcal{Y}(\tau_0^{(3)}, \tau_1^{(3)})$ by forming

$$\mathcal{Q}^{(3)} = \mathcal{N}_{\text{cubic}}^{(3)} - \kappa_3 \mathcal{Y}(\tau_0^{(3)}, \tau_1^{(3)}). \quad (25)$$

The trilinear kernel criterion requires that the projection of $\mathcal{Q}^{(3)}$ onto the second-derivative sector vanish. Concretely, we impose

$$\begin{aligned} \left[\mathcal{Q}^{(3)} \right]_{\partial_\xi^2 \tau_0^{(3)}} &= 0, & \left[\mathcal{Q}^{(3)} \right]_{\partial_\xi^2 \tau_1^{(3)}} &= 0, \\ \left[\mathcal{Q}^{(3)} \right]_{\partial_\eta^2 \tau_0^{(3)}} &= 0, & \left[\mathcal{Q}^{(3)} \right]_{\partial_\eta^2 \tau_1^{(3)}} &= 0. \end{aligned} \quad (26)$$

Equivalently, all coefficients of

$$\partial_\xi^2 \tau_0^{(3)}, \quad \partial_\xi^2 \tau_1^{(3)}, \quad \partial_\eta^2 \tau_0^{(3)}, \quad \partial_\eta^2 \tau_1^{(3)} \quad (27)$$

in $\mathcal{Q}^{(3)}$ must vanish simultaneously.

Carrying out this coefficient comparison gives a single consistent value of the normalization constant κ_3 . With this value, all second-derivative terms cancel identically:

$$\mathcal{Q}^{(3)} = \mathcal{N}_{\leq 1}^{(3)}, \quad (28)$$

where $\mathcal{N}_{\leq 1}^{(3)}$ contains only first derivatives of the tau functions.

Thus the $\delta = 3$ Tomimatsu–Sato solution satisfies

$$\mathcal{N}_{\text{cubic}}^{(3)} = \kappa_3 \mathcal{Y}(\tau_0^{(3)}, \tau_1^{(3)}) + \mathcal{N}_{\leq 1}^{(3)}. \quad (29)$$

This proves that the second-derivative sector of the $\delta = 3$ TS solution is governed by the same YTSF-type trilinear kernel as in the $\delta = 2$ case. This implies that any *delta*-dependence can only enter through the normalization factor κ . The explicit coefficient extraction and the determination of κ_3 are summarized in Appendix A, while the computational implementation of this procedure is given in Appendix B. Substituting the explicit determinant tau functions given in Eq. (23), we have explicitly verified that all second-derivative coefficients vanish identically.

5 Universality of the Trilinear Kernel in the TS Hierarchy

The results obtained for the $\delta = 2$ and $\delta = 3$ cases strongly suggest that the trilinear kernel structure identified in Sec. 2 is not accidental, but extends to higher members of the Tomimatsu–Sato (TS) hierarchy.

A key observation is that the cubic part of the Ernst numerator,

$$\mathcal{N}_{\text{cubic}} = \tau_1 \tau_0^2 \Delta \tau_1 - \tau_1^2 \tau_0 \Delta \tau_0, \quad (30)$$

is determined solely by the tau-ratio structure

$$\mathcal{E} = \frac{\tau_1}{\tau_0}, \quad (31)$$

and does not depend on the explicit form of the tau functions.

Therefore, for any δ , the second-derivative structure of the Ernst equation is universal at the level of $\mathcal{N}_{\text{cubic}}$. The only nontrivial question is whether this universal structure can always be matched by a trilinear kernel of YTSF type.

The results of Sec. 4 show that this matching holds for $\delta = 3$. Combined with the previously established $\delta = 2$ case, this leads to the following conjecture.

Universality conjecture. For the entire Tomimatsu–Sato hierarchy,

$$\mathcal{N}_{\text{cubic}} = \kappa(\delta) \mathcal{Y}(\tau_0, \tau_1) + \mathcal{N}_{\leq 1}(\tau_0, \tau_1), \quad (32)$$

where $\mathcal{N}_{\leq 1}$ contains only first derivatives of the tau functions.

If true, this implies that the stationary axisymmetric Einstein equations, when expressed in tau-ratio form, possess a universal trilinear kernel governing their second-order structure, while all higher nonlinearities are encoded in a lower-derivative envelope.

This suggests that the TS hierarchy realizes a two-layer integrable structure:

$$\text{Toda (bilinear)} \quad + \quad \text{Ernst (trilinear)}. \quad (33)$$

This follows from the fact that $\mathcal{N}_{\text{cubic}}$ is entirely determined by the tau-ratio structure and does not depend on the detailed form of the tau functions. Establishing this conjecture for general δ , as well as extending it to other stationary axisymmetric solutions, remains an important direction for future work.

6 Discussion: From Bilinear to Trilinear Dynamics

We conclude by discussing the physical meaning of the transition from bilinear to trilinear structures.

In the standard Hirota formalism, bilinear equations describe nonlinear dynamics in terms of pairwise tau-function relations. Their basic building block is of the form

$$\tau \partial^2 \tau - (\partial \tau)^2, \quad (34)$$

or, equivalently, Hirota derivatives acting on two copies of a tau function,

$$D_x^n \tau \cdot \tau. \quad (35)$$

This structure is naturally associated with interference phenomena, pairwise interactions, and the Grassmannian picture of Sato theory.

The trilinear case represents a qualitatively different level of nonlinearity. Its characteristic highest-derivative structure is of the schematic form

$$\tau^2 \partial^2 \tau, \quad (36)$$

or, more generally, a three-slot coupling

$$T(\tau_a, \tau_b, \tau_c). \quad (37)$$

Thus trilinearity should not be viewed as a simple algebraic extension of bilinearity, but as a distinct dynamical structure in which the fundamental nonlinear kernel involves three tau functions.

This distinction becomes particularly transparent in the Ernst system. As shown in Secs. 3–4, writing the Ernst potential in tau-ratio form,

$$\mathcal{E} = \frac{\tau_1}{\tau_0}, \quad (38)$$

the numerator of the equation separates as

$$\mathcal{N} = \mathcal{N}_{\text{cubic}} + \mathcal{N}_{\text{quartic}}, \quad (39)$$

where all second-derivative terms are contained in $\mathcal{N}_{\text{cubic}}$, while $\mathcal{N}_{\text{quartic}}$ consists only of first-derivative products. Therefore the dynamical kernel of the equation is trilinear, while the quartic part acts only as a lower-derivative envelope.

This observation suggests a general principle. For a second-order field equation written in tau-ratio form, the highest-derivative sector can contain at most cubic tau-function combinations. Quartic or higher multilinear structures may appear algebraically, but they do not control the highest differential order unless the underlying field equation itself contains higher derivatives. Higher-derivative field theories are known to suffer from the Ostrogradsky instability, which leads to ghost degrees of freedom [12, 13] and threatens the unitarity of the quantum theory [14, 15]. In particular, higher-derivative gravity theories, although renormalizable, typically contain massive ghost modes.

In the present framework, this restriction manifests itself in the fact that the highest-derivative sector of the tau-function representation is necessarily cubic, leading to a trilinear integrable kernel. In this sense, the trilinear structure represents the maximal nontrivial integrable kernel compatible with a second-order dynamical equation. This is particularly natural in relativistic field theory, where higher-derivative equations typically lead to instabilities or acausal (or non-unitary) modes. The emergence of a trilinear kernel can therefore be viewed as a reflection of the second-order and causal character of the underlying dynamics.

The transition

$$\text{bilinear} \quad \longrightarrow \quad \text{trilinear} \quad (40)$$

thus admits a physical interpretation. Bilinear equations describe integrable systems organized by pairwise tau-function relations, whereas trilinear equations describe systems in which the field participates more directly in the formation of its own nonlinear background. In gravitational systems this is especially natural, since the field itself determines the geometry in which it propagates.

This leads to the following working viewpoint:

$$\boxed{\text{second-order self-interacting geometry} \quad \Rightarrow \quad \text{trilinear tau-function kernel}}. \quad (41)$$

The Tomimatsu–Sato realization of the YTSF-type kernel provides a concrete example of this principle. Extending this viewpoint to other stationary axisymmetric solutions, and possibly to more general relativistic field theories, remains an interesting direction for future work.

Appendices

A Coefficient Extraction for the $\delta = 3$ Case

In this appendix we present the coefficient-level analysis used in Sec. 4.

Starting from

$$\mathcal{Q}^{(3)} = \mathcal{N}_{\text{cubic}}^{(3)} - \kappa_3 \mathcal{Y}(\tau_0^{(3)}, \tau_1^{(3)}), \quad (42)$$

we extract the coefficients of the independent second derivatives

$$\partial_\xi^2 \tau_0^{(3)}, \quad \partial_\xi^2 \tau_1^{(3)}, \quad \partial_\eta^2 \tau_0^{(3)}, \quad \partial_\eta^2 \tau_1^{(3)}. \quad (43)$$

We denote these coefficients by

$$C_{0\xi}, \quad C_{1\xi}, \quad C_{0\eta}, \quad C_{1\eta}. \quad (44)$$

As an example, the coefficient of $\partial_\xi^2 \tau_1^{(3)}$ takes the form

$$C_{1\xi} = \tau_0^{(3)2} \tau_1^{(3)} - \kappa_3 F_{1\xi}(\tau_0^{(3)}, \tau_1^{(3)}, \partial\tau). \quad (45)$$

Here $F_{1\xi}$ is a polynomial involving τ and first derivatives and the leading second-derivative structure is explicitly given by the $\tau_0^2 \tau_1$ term.

The remaining coefficients $C_{0\xi}, C_{0\eta}, C_{1\eta}$ have analogous structures.

Substituting the explicit $\delta = 3$ tau functions and solving

$$C_{0\xi} = C_{1\xi} = C_{0\eta} = C_{1\eta} = 0 \quad (46)$$

determines κ_3 . The coefficient comparison described above yields (See Appendix B)

$$\kappa_3 = -4p^2 q^2, \quad (47)$$

which coincides with the result obtained in the $\delta = 2$ case. Here the parameters p and q are dimensionless constants characterizing the Tomimatsu–Sato solution, satisfying

$$p^2 + q^2 = 1. \quad (48)$$

They are related to the mass M and angular momentum parameter $a = J/M$ through

$$p = \frac{\sigma}{M}, \quad q = \frac{a}{M}, \quad (49)$$

where $\sigma^2 = M^2 - a^2$. This indicates that the normalization of the trilinear kernel is independent of δ , suggesting that the trilinear structure is a property of the Ernst equation itself. With this value, all second-derivative terms cancel identically:

$$\mathcal{Q}^{(3)} = \mathcal{N}_{\leq 1}^{(3)}. \quad (50)$$

B Mathematica Verification

The trilinear kernel criterion can be efficiently checked using *Mathematica* by projecting onto the second-derivative sector.

A minimal implementation is as follows:

```
ClearAll["Global`*"];
```

```
tau0 = tau0[xi, eta];
```

```
tau1 = tau1[xi, eta];
```

```
omega = Exp[2 Pi I/3];
```

```
Tx[f_, g_, h_, x_] :=
```



```

Module[{x1, x2, x3},
  (D[f /. x -> x1, x1] (g /. x -> x2) (h /. x -> x3)
   + omega (f /. x -> x1) D[g /. x -> x2, x2] (h /. x -> x3)
   + omega^2 (f /. x -> x1) (g /. x -> x2)
   D[h /. x -> x3, x3])
  /. {x1 -> x, x2 -> x, x3 -> x}
];

Y =
D[tau1 Tx[tau0, tau0, tau1, xi]
 - tau0 Tx[tau1, tau1, tau0, xi], xi]
+
D[tau1 Tx[tau0, tau0, tau1, eta]
 - tau0 Tx[tau1, tau1, tau0, eta], eta];

Ncubic =
tau1 tau0^2 (D[tau1, {xi, 2}] + D[tau1, {eta, 2}])
-
tau1^2 tau0 (D[tau0, {xi, 2}] + D[tau0, {eta, 2}]);

Q = Expand[Ncubic - k Y];

coeffs = {
Coefficient[Q, D[tau0, {xi, 2}]],
Coefficient[Q, D[tau1, {xi, 2}]],
Coefficient[Q, D[tau0, {eta, 2}]],
Coefficient[Q, D[tau1, {eta, 2}]]
};

Solve[coeffs == {0, 0, 0, 0}, k]

```

Substituting the explicit $\delta = 3$ tau functions into this procedure confirms that the second-derivative terms cancel identically. This procedure directly implements the coefficient-level projection described in Appendix A. The solution of the above system is

$$k = -4. \quad (51)$$

Substituting this value into Q , we confirm that

$$Q = 0, \quad (52)$$

i.e., all second-derivative terms cancel identically.

Restoring the normalization used in Appendix A, we obtain Eq. (47).

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